

M35 — Master Start/Pause/End controls + instructor auto-stations + engage deep-link

Phase: Phase 7 follow-on (post-M34, operator-feedback fix) **Status:** **DONE** **Blocked by:** M2 (Encounter dataclass), M4 (room API), M5 (control_room dashboard), M22 (Per-Patient Console), M33 (Engage button)

Blocks: none **Estimated effort:** 1 day

1. Purpose

Operator feedback after M34:

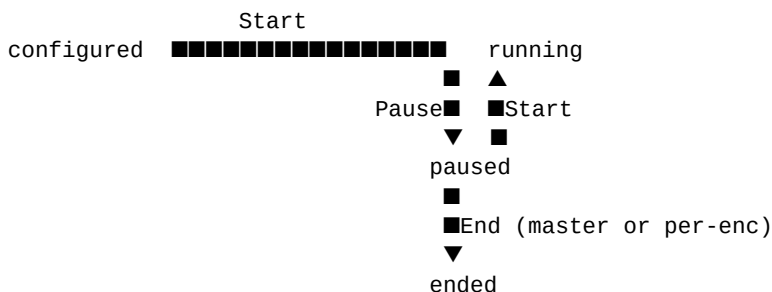
"On encounter control need to have in the header, start scenario (launching all scenarios at the same time) pause (pause all scenarios) and end (end/stop all scenarios and launch debrief) and in the individual encounter have the same headers structure but only impacting the specific encounter. For the stop the debrief is not called up until the main control page for encounter has stopped all the debriefs. Also for the instructor engagement in an encounter the instructor has an engagement button, this should not launch Join a MEDSIM 2 session this should happen behind the scene when the scenario is started by the master control in the main encounter page."

Four asks bundled into one module:

1. **Master Start / Pause / End on** /portal/room. Three buttons in the Multi-Patient Control header. Start launches every encounter simultaneously. Pause pauses everything. End stops everything *and* fires the cohort debrief (existing behavior — just renamed).
2. **Per-encounter Start / Pause / End on** /portal/room/encounter/{id}. Same shape, scoped to one bed.
3. **Per-encounter End does NOT save the cohort debrief.** That only happens on master End. Operator can wrap up beds one-by-one as students finish, then hit master End once to save the combined PEARLS debrief at the point all beds are done.
4. **Engage skips the public /join page entirely.** Master Start auto-registers an INST-<persona_id> chat station for every persona on every encounter. The Engage button on the Per-Patient Console voice card now deep-links straight into that station, so the instructor lands on the chat UI without typing a name or picking a persona.

2. Structure

2.1 State machine



configured is the dataclass default for a freshly-created encounter (after POST /api/room/start). Master Start (or per-encounter Start) is the trigger that moves it into running.

2.2 Files touched

- portal/ws_room.py — two new emitters: `emit_start_all(room_code,

encounter_count) and emit_encounter_state(room_code, encounter_id, state)`.

- portal/server.py — five new routes + two helpers:
- POST /api/room/start_all — master Start.
- POST /api/encounter/{id}/start — per-encounter Start.
- POST /api/encounter/{id}/pause — per-encounter Pause.
- POST /api/encounter/{id}/end — per-encounter End (no debrief).
- GET /portal/engage/{encounter_id}/{persona_id} — instructor

engage deep-link. - _instructor_station_id_for(persona_id) — deterministic "INST-{pid}" station id (so lookups are O(1), no scans). - _ensure_instructor_stations(enc) — for each persona in enc.selected_personas, ensure a chat station exists with id INST-<pid> (idempotent).

- portal/templates/control_room.html — header rewired: btn-start-all

(new) + btn-freeze (relabeled "■ Pause all") + btn-end (relabeled "■ End all (debrief)") + existing btn-scene / btn-debrief. btn-resume removed from the visible header (Start handles both first-launch and resume-after-pause).

- portal/static/control_room.js — wires btn-start-all to

/api/room/start_all. Master End now follows the cohort_debrief_url in the response — instructor lands on the PEARLS debrief immediately, no manual click.

- portal/templates/encounter_console.html — three new buttons in

.console-header-right: btn-enc-start (primary), btn-enc-pause (secondary), btn-enc-end (danger, with confirm dialog explaining the no-debrief contract).

- portal/static/encounter_console.js — wires the three new buttons

and **rewrites the Engage anchor** href in bootVoices() from /join?code={joinCode} to /portal/engage/{encounterId}/{persona_id}.

2.3 Instructor station shape

The instructor station is a regular Station dataclass row in enc.stations, distinguished by two markers:

Field	Value
station_id	INST-{persona_id} (deterministic, prefix INST-)
persona_id	the persona this station plays
user_agent	"instructor-engage" (informational tag)

No schema change. The Station table is in-memory only (matches the M2 contract — chat stations live as long as the singleton).

3. Uses

3.1 Master control flow

1. Instructor finalizes the wizard → /api/room/start creates the room with N encounters all in state configured. 2. Instructor lands on /portal/room. Header shows *■ Start all scenarios · ■ Pause all · ■ Inject scene · ■ Cohort debrief · ■ End all (debrief)*. 3. Click **■ Start all scenarios** → POST /api/room/start_all: - Every encounter's state becomes running (skipped if already ended). - For

every persona in every encounter, an INST-`<pid>` station is created. - WS broadcast `start_all` so subscribed stations transition simultaneously. - Response: `{ok, status, encounter_count, instructor_stations_created}`. 4. Click **Pause all** → existing POST `/api/room/freeze_all`. 5. Click **End all (debrief)** → existing POST `/api/room/end` — saves cohort debrief, clears singleton, and (M35) the JS handler redirects directly to `cohort_debrief_url` in the response.

3.2 Per-encounter control flow

1. Instructor opens `/portal/room/encounter/{id}` (Per-Patient Console). 2. Header shows **Start · Pause · End · Open EHR · Pop out**. 3. Click **Start** → POST `/api/encounter/{id}/start`: - This encounter's state becomes `running`. - Instructor stations created for its personas. - State badge in the header updates live. 4. Click **Pause** → POST `/api/encounter/{id}/pause`. 5. Click **End** → confirm dialog: `"End this encounter? The bed will be marked ended but the cohort debrief is NOT saved until you press 'End all (debrief)' on the master control."` → POST `/api/encounter/{id}/end` → state becomes `ended`, NO cohort debrief is saved, singleton stays alive.

3.3 Engage flow

1. Instructor opens Per-Patient Console → voice card lists every persona. 2. Click **Engage** on (say) the `*Anxious Parent (P-015)*` row. 3. Browser navigates to `/portal/engage/{encounter_id}/P-015` (new tab via `target="_blank"`). 4. Server resolves the encounter, ensures INST-P-015 exists in `enc.stations` (lazy-create if master Start hasn't fired yet), 303 redirects to `/station/{join_code}/INST-P-015`. 5. Instructor lands on the standard chat-station UI bound to the parent persona. No `/join` handshake. No name typed. No persona picker.

4. Functions (exported API surface)

Symbol	Where	Purpose
POST <code>/api/room/start_all</code>	<code>portal/server.py</code>	Master Start.
POST <code>/api/encounter/{id}/start</code>	<code>portal/server.py</code>	Per-encounter Start.
POST <code>/api/encounter/{id}/pause</code>	<code>portal/server.py</code>	Per-encounter Pause.
POST <code>/api/encounter/{id}/end</code>	<code>portal/server.py</code>	Per-encounter End (no debrief).
GET <code>/portal/engage/{eid}/{pid}</code>	<code>portal/server.py</code>	303 → <code>/station/{join}/INST-{pid}</code> . Lazy-creates the station.
<code>_instructor_station_id_for(pid)</code>	<code>portal/server.py</code>	Returns <code>"INST-{pid}"</code> .
<code>_ensure_instructor_stations(enc)</code>	<code>portal/server.py</code>	Idempotent helper used by all Start paths.
<code>emit_start_all(room_code, encounter_count)</code>	<code>portal/ws_room.py</code>	Broadcasts <code>{type: "start_all"}</code> on the room channel.
<code>emit_encounter_state(room_code, encounter_id, state)</code>	<code>portal/ws_room.py</code>	Broadcasts <code>{type: "encounter_state", encounter_id, payload: {state}}</code> .

5. Limitations

- ****Per-encounter Pause does not stop the bedside chat UI from**

appearing to accept input.** The student's chat station polls state every 2s and reacts then; sub-poll-cycle latency is the same as M16's freeze_all behavior. WS push exists (emit_encounter_state) but bedside JS does not yet subscribe to it — that's a future M36.

- **Master Start is idempotent but does not re-broadcast** a fresh

start_all WS message on every click. If an instructor clicks Start, then Start again, only the first click fires WS (the second click still calls the route and returns OK, but most subscribers won't see anything new).

- **The instructor station has no chat history at lazy-create time.**

If an instructor clicks Engage before master Start, the persona may not have warmed-up state from a previous turn. Acceptable — this is the safety-net path; the normal flow is master Start first.

- ****INST-{persona_id} collides with a student-typed station_id**

if a real student deliberately picked an id starting with INST-**. Student ids are generated server-side (secrets.token_urlsafe) so this won't happen in practice; documented for future-proofing.

- ****Per-encounter End sets state="ended" but does NOT remove the**

encounter from the room.** The dashboard still shows it; the badge reads ENDED. Master End is still the path that nukes the singleton + saves the debrief. This is the user-explicit contract.

- ****The btn-resume button was removed from the master header**

visually but its JS handler remains** (defensive — cached stale DOM won't error). The /api/room/resume_all route also stays for v6-compat scripts and the per-encounter Start path effectively replaces it.

- ****The cohort debrief auto-redirect on master End runs from the**

control_room.js handler.** If the JS doesn't run (e.g. instructor hit End from a non-JS client), the response body still contains cohort_debrief_url so manual navigation works.

6. Test status

Test file	Asserts	Status	Last run
tests/v7/test_master_controls_and_engage.py::test_master_start_all_transitions_every_encounter_to_running	All beds → running; INST-<pid> stations created for every persona	PASS	2026-05-27
...::test_master_start_all_is_idempotent	Second Start creates 0 new stations	PASS	2026-05-27
...::test_pause_all_after_start_all_marks_every_encounter_paused	After Start then freeze_all every state = paused	PASS	2026-05-27
...::test_master_end_saves_cohort_debrief_and_clears_singleton	Master End contract intact	PASS	2026-05-27

Test file	Asserts	Status	Last run
...::test_per_encounter_start_sets_running_and_creates_instructor_stations	One-bed Start scoped correctly; other beds untouched	PASS	2026-05-27
...::test_per_encounter_pause_sets_paused	One-bed Pause scoped correctly	PASS	2026-05-27
...::test_per_encounter_end_marks_ended_but_NO_cohort_debrief	Per-encounter End → state=ended, NO cohort file, singleton alive	PASS	2026-05-27
...::test_per_encounter_routes_404_unknown_encounter	Unknown encounter id → 404 on all three actions	PASS	2026-05-27
...::test_engage_redirects_to_instructor_station_after_master_start	After Start, engage → 303 to /station/{join} /INST-{pid}	PASS	2026-05-27
...::test_engage_lazy_creates_station_before_master_start	Engage works pre-Start; station lazy-created	PASS	2026-05-27
...::test_engage_rejects_persona_not_on_encounter	Engage with wrong persona → 404	PASS	2026-05-27
...::test_multi_patient_control_header_has_start_pause_end_buttons	Master header has Start all / Pause all / End all (debrief)	PASS	2026-05-27
...::test_per_patient_console_header_has_start_pause_end_buttons	Console header has btn-enc-start / pause / end + confirm copy	PASS	2026-05-27
...::test_encounter_console_js_uses_portal_engage_url_not_join	Engage anchor href uses /portal/engage/, not /join?code=	PASS	2026-05-27
(regression update) ...test_encounter_console_js_renders_name_test_and_engage	Asserts updated from /join?code= to /portal/engage/	PASS	2026-05-27
Full v7 suite	231 passed, 1 skipped (M20 Playwright skip — unchanged)	PASS	2026-05-27

7. Change list

Date	Author	Change	Files
2026-05-27	claude-code	Initial M35 implementation: 5 new routes + 2 WS emitters + 2 helpers; master header rewired (Start all / Pause all / End all (debrief)); per-encounter header gets Start / Pause / End; engage URL flipped from /join to /portal/engage; 14 new tests; 1 phrase-only regression update	portal/ws_room.py, portal/server.py, portal/templates/control_room.html, portal/static/control_room.js, portal/templates/encounter_console.html, portal/static/encounter_console.js, tests/v7/test_master_controls_and_engage.py (new), tests/v7/test_encounter_characters_voices_engage.py (assertion text only)

8. Open questions / known issues

• Should bedside stations auto-disable input on per-encounter Pause?

Currently the 2s state poll is the latency. M16 has `emit_freeze_all` for master pause; M35 added `emit_encounter_state` but bedside JS doesn't subscribe yet. Tracked as a future M36.

• **Should master Start prevent students who haven't joined yet from

joining mid-scenario?** Currently no — students can `/join?code=X` at any state. Acceptable for v7.0; we may add a "lock roster" flag on Start later.

• Cohort debrief auto-redirect could be opt-in. Some instructors

may want to stay on the dashboard after End to discuss before reviewing the debrief. Trivial follow-up: a "Skip debrief" confirm variant.

• Per-encounter Start when the encounter is ended is a no-op

(returns success but doesn't change state). We may want to reject it with a 409 to surface the bug-vs-intentional ambiguity. Out of scope today.

*Render this guide to PDF with `python docs/module_guides/render_pdfs.py.*`